Divisor Channels and Two Arithmetic Bridges

A proof audit and rational continuation of the Koide–Erdős–Straus packet

Codex reconstruction and verification of supplied Pro-mode calculations

5 September 2026 — research draft

Abstract

We verify the complete oriented divisor-channel map, the diagonal-ray certificate count, and its classical density-one argument. The integral Bézout opening is retained with its precise positivity hypothesis and explicit integral presentation maps. The supplied orthogonal map to the defect-one Koide surface has no rational image point. We give a different rational map, its inverse on specified open sets, and its exact denominator formula. This extends the arithmetic comparison without asserting a global threefold realization, a particle model, or a proof of Erdős–Straus. For $p = 1201$, the complete fifteen-state selector census maps to fifteen distinct projective rays in the upper-left real symmetric corner of $\mathrm{Herm}_3(\mathbb{H})$. Their linear span and Jordan closure both have dimension three. Exact counts $15, 54, 36, 29$ at four primes in the same residue class show that fifteen is not a residue-class invariant.

1 Divisor channels, density one, and two distinct arithmetic maps

This dated continuation (4 September 2026) checks the supplied Pro-mode *Divisor Channels, Prime Saturation, and Bézout Openings*, version 3, and extends its final arithmetic comparison. The original packet is retained unchanged under `supporting_materials/workbench/evidence/koide_es_v3_audit_20260904/source/`. The claims below concern explicit reciprocal equations, integral matrices, and quadratic surfaces. They do not require a complex structure on a sphere.

1.1 The complete oriented channel map

Fix integers $n \equiv 1 \pmod 4$, $R \equiv 3 \pmod 4$, $0 < R < n$, $\gcd(n, R) = 1$, and put $a = (n + R)/4$, $N = na$. Then $\gcd(n, a) = \gcd(R, a) = 1$. For a positive divisor $\delta \mid n^2$ define

$$\mathcal{C}_\delta(n, R) = \{(u, v) \in \mathbb{Z}^2_{>0} : uv = a^2,\ R \mid 4u + \delta\}.$$

Proposition 1.1. *The following is a bijection onto all ordered positive solutions of* $4/n = 1/a + 1/y + 1/z$*:*

$$(\delta, u, v) \longmapsto (y, z) = \left(\frac{N + (n^2/\delta)u}{R}, \frac{N + \delta v}{R}\right), \qquad (u, v) \in \mathcal{C}_\delta(n, R).$$

Its inverse, retaining this orientation, is

$$d = Ry - N, \quad e = Rz - N, \quad k = \gcd(d, n^2), \quad \delta = n^2/k, \quad u = d/k, \quad v = e/\delta.$$

Exchange of y, z *is the fixed-point-free involution* $(\delta, u, v) \mapsto (n^2/\delta, v, u)$.

Proof. The reciprocal equation is $Ryz = N(y+z)$, hence $de = N^2$. For any positive solution $Ry = N + Ny/z > N$, so $d > 0$; similarly $e > 0$. Because n and a are coprime, every prime dividing n occurs in de with precisely its exponent in n^2. Thus k extracts the entire n-primary part of d, δ the entire n-primary part of e, and u, v are positive integers with $uv = a^2$. Reducing $N + e$ modulo R gives $0 \equiv na + \delta v \equiv 4a^2 + \delta v = v(4u + \delta)$. Since v is a unit modulo R, this proves the channel condition. Conversely that condition makes $N + e$ divisible by R; also $\delta(N + d) = n(\delta a + nu) \equiv na(\delta + 4u) \equiv 0 \pmod{R}$. The unit δ can be cancelled modulo R, proving integrality of y. The identity $de = N^2$ proves the reciprocal equation. These operations are inverse. A fixed point of the involution would have $\delta = n$, $u = v = a$, so $R \mid 4a + n = 2n + R$, impossible since $R \geq 3$ is odd and coprime to n. □

The package therefore corrects the earlier orientation reversal: the channel order is $n^2/\gcd(Ry - na, n^2)$, not $\gcd(Ry - na, n^2)$. For $n = 9$, $R = 7$, $a = 4$, $(\delta, u, v) = (3, 1, 16)$, the ordered solution is $(y, z) = (9, 12)$, and $\gcd(27, 81) = 27$.

1.2 The integral presentation, with its exact realization boundary

Put $T = (4u + \delta)/R$, $j = (R - 3)/4$, and $\sigma = 1$ for $\delta \equiv 1 \pmod 4$, $\sigma = -1$ otherwise. Define

$$\ell_0 = (T + \sigma)/4, \quad \ell_1 = u - jT = (3T - \delta)/4, \quad \ell_2 = \sigma, \quad \mathsf{R}_\delta = \begin{pmatrix} 3 & -\ell_1 \\ 4 & -T \end{pmatrix}.$$

All coordinates are integral because $T \equiv -\delta \pmod 4$. Direct multiplication and subtraction give

$$\begin{pmatrix} 1 & -j \\ 0 & 1 \end{pmatrix} \begin{pmatrix} R & -u \\ 4 & -T \end{pmatrix} = \mathsf{R}_\delta, \qquad 12\ell_0 - 4\ell_1 - 3\ell_2 = \delta.$$

The entries $3, 4$ are coprime, so $\mathrm{SNF}(\mathsf{R}_\delta) = \mathrm{diag}(1, \delta)$; in particular its cokernel is $\mathbb{Z}/\delta\mathbb{Z}$. Reducing $4\ell_1 = 3T - \delta$ modulo three gives $\ell_1 \equiv -\delta \pmod 3$. Thus the source gluing admissibility $3 \nmid \ell_1$ fails precisely when $3 \mid \delta$. For the example above $(\ell_0, \ell_1, \ell_2) = (0, 0, -1)$. The matrix and its cokernel remain valid, but assigning the source's global manifold homology to this nonadmissible triple is not established by this calculation. This is an explicit admissibility test, not a failure of the reciprocal solution or the integral matrix morphism.

1.3 An exact density-one statement and what it leaves out

Proposition 1.2. *Let $p \equiv 1 \pmod 4$ be prime and $m = (p + 1)/2$. The exterior channel states on $u = a = v$ are indexed exactly by*

$$\{R : R \mid m,\ R \equiv 3 \pmod 4\}.$$

For each such R, set $x = (p + R)/4$ and $k = (p + 1)/R$. Then

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{xk} + \frac{1}{pxk}, \qquad \nu(p) = \frac{1}{2}\left(\tau(m) - \sum_{d \mid m} \chi_4(d)\right),$$

where $\chi_4(d) = 1$ for $d \equiv 1 \pmod 4$ and -1 for $d \equiv 3 \pmod 4$. The count is zero exactly when every prime divisor of m is $1 \pmod 4$.

Proof. For $\delta = 1$ and $u = a$, the condition is $R \mid 4a + 1 = p + R + 1$, equivalently $R \mid p + 1$. Since R is odd this is $R \mid m$. Every such R obeys $3 \le R \le (p+1)/2 < p$, and $k \ge 2$. Adding the final two fractions gives $(p+1)/(pxk) = R/(px)$, and adding $1/x$ gives $(p+R)/(px) = 4/p$. Distinct R give distinct x. Summing the indicator $(1 - \chi_4(d))/2$ over divisors of m proves the count. A divisor of residue three exists exactly when a prime factor of residue three exists: the converse follows because products of residue-one primes and their powers have residue one. □

Proposition 1.3. *For fixed $4 \mid M$, $\gcd(c, M) = 1$, $c \equiv 1 \pmod 4$, the preceding diagonal states cover relative density one of primes $p \equiv c \pmod M$. For a finite set $\mathcal{Q}$ of primes $q \equiv 3 \pmod 4$, $q \nmid M$, the subfamily with some $q \mid p + 1$ has exact relative density*

$$1 - \prod_{q \in \mathcal{Q}} \frac{q-2}{q-1}.$$

Proof. The Chinese remainder theorem gives $\prod(q-1)$ reduced residue classes modulo $M \prod q$ above the specified class modulo M. Avoiding $p \equiv -1 \pmod q$ at every q leaves exactly $\prod(q-2)$ of them. The prime number theorem for each fixed arithmetic progression gives the displayed ratio. For completeness, its modulus-four case and partial summation give

$$\sum_{\substack{q \le Y \\ q \equiv 3 \ (4)}} \frac{1}{q} = \frac{\pi(Y;4,3)}{Y} + \int_2^Y \frac{\pi(t;4,3)}{t^2}\, dt \sim \tfrac{1}{2} \log\log Y.$$

Removing finitely many primes dividing M does not change divergence. Since $\log(1-z) \le -z$ for $0 < z < 1$, the finite products tend to zero. The full family's lower relative density is at least each of these finite union densities, hence is at least one and therefore equals one. No interchange of two unproved asymptotic limits is needed. □

These analytic tools are classical. Elsholtz and Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, J. Aust. Math. Soc. 94 (2013), 50–105, arXiv:1107.1010v6, Appendix A, Lemma A.3, give a stronger shifted-prime additive-function statement, attributed there to Barban via Elliott. Choosing its prime weight to be the indicator of residue three gives the relevant abundance of factors. The finite-CRT proof above records the precise chosen ray and progression; it is not a claim that almost-all solvability is new.

For $p = 73$, $(p+1)/2 = 37$ has no residue-three prime factor. This prime therefore misses the diagonal ray, although

$$\frac{4}{73} = \frac{1}{20} + \frac{1}{220} + \frac{1}{4015}.$$

Here $R = 7$, $u = 5$, $v = 80$, and the primitive factor ray is $[1 : 4]$. Thus a prime missed by the selected ray need not be missed by the full selector. Density one does not settle the universal Erdős–Straus claim.

1.4 The Bézout opening, with the positivity hypothesis retained

For coprime positive integers r, t, choose $x, y \in \mathbb{Z}$ with $ry - tx = 1$, set $v = (r,t)^{\mathsf{T}}$, $w = (x,y)^{\mathsf{T}}$, and let

$$B = (v\ w) \in \mathrm{SL}_2(\mathbb{Z}), \qquad A = \delta v v^{\mathsf{T}} - w w^{\mathsf{T}} = B\,\mathrm{diag}(\delta, -1) B^{\mathsf{T}}.$$

Then $\det A = -\delta$ for every real δ. All companions are $w + kv$ for $k \in \mathbb{Z}$: subtraction of two Bézout equations gives $r\Delta y = t\Delta x$, so coprimality gives $(\Delta x, \Delta y) = k(r,t)$. Choosing $|x| \le r/2$ retains an

actual companion rather than quotienting away its dependence. For a positive integer channel order,

$$A_{11} = \delta r^2 - x^2 \geq (\delta - 1/4)r^2 > 0.$$

The broader wording "every $\delta > 0$" does not imply that inequality. At $(r, t, \delta) = (2, 1, 1/8)$ all companions have x odd, and $A_{11} = 1/2 - x^2 \leq -1/2$. For a specified real order and companion the exact necessary and sufficient condition is $\delta > (x/r)^2$. This correction leaves every positive integral divisor channel intact.

With $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$ and $a > 0$, the periods $(\tau, \mu, \beta) = (ia, ib/\sqrt{6}, ic)$ give

$$P_A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \sqrt{6}b & a & 0 & 0 \\ 0 & 0 & 0 & 1 \\ c & b/\sqrt{6} & 0 & 0 \end{pmatrix}, \qquad \det P_A = ac - b^2 = -\delta.$$

This invertible real matrix sends $\mathbb{Z}^4$ to a discrete cocompact lattice in $\mathbb{C}^2$ and therefore defines a complex two-torus. The determinant sign and the factor $\sqrt{6}$ are retained. It does not determine a point on the particular global equivariant period curve of the proposed threefold. That incidence problem remains a distinct map problem, not an obstruction to the torus just constructed.

There is also an explicit integral equivalence of presentations, not merely an equality of determinants. Set

$$P_0 = \begin{pmatrix} -1 & 1 \\ -4 & 3 \end{pmatrix}, \quad Q_0 = \begin{pmatrix} 1 & T - \ell_1 \\ 0 & 1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Multiplication gives $P_0 \mathsf{R}_\delta Q_0 = \operatorname{diag}(1, -\delta)$ and $(-JB^{-1})A(B^{-\mathsf{T}}J) = \operatorname{diag}(1, -\delta)$. Thus, for $L = P_0^{-1}(-JB^{-1})$, $V = B^{-\mathsf{T}}JQ_0^{-1}$,

$$\mathsf{R}_\delta = LAV.$$

As maps $\mathbb{Z}^2 \to \mathbb{Z}^2$, the commuting square has source isomorphism V^{-1} and target isomorphism L because $\mathsf{R}_\delta V^{-1} = LA$. Hence $[z] \mapsto [Lz]$ is an explicit isomorphism of the two cokernels. It is not by itself a map of tori or of global threefolds.

1.5 The original map's rational nonintersection

Retain the exact form and orthogonal matrix

$$\Delta(q) = 4(q_1q_2 + q_1q_3 + q_2q_3) - q_1^2 - q_2^2 - q_3^2, \quad F = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \end{pmatrix}.$$

For $\det A = -\delta$, $\delta \in \mathbb{Q}_{>0}$, Pro's map is

$$\mathfrak{K}_\delta(A) = F^{-1} \frac{((a + c)/2, b, (a - c)/2)^{\mathsf{T}}}{\sqrt{3\delta}}.$$

It satisfies $\Delta(\mathfrak{K}_\delta(A)) = -1$ but sends no rational A to a rational triple. Indeed, if q were rational, put $S = q_1 + q_2 + q_3$, $U = q_1 - q_3$, $V = q_1 + q_3 - 2q_2$. The rational coordinates $H = (a + c)/2$, $X = b$, $Y = (a - c)/2$ would satisfy

$$H = \sqrt{\delta}S, \quad X = \sqrt{3\delta/2}\,U, \quad Y = \sqrt{\delta/2}\,V.$$

Two nonzero coordinates would make one of $\sqrt{3/2}$, $\sqrt{2}$, $\sqrt{3}$ rational. If only H is nonzero the determinant is positive. If only X or only Y is nonzero, the determinant equation forces, respectively, $U^2 = 2/3$ or $V^2 = 2$, impossible over $\mathbb{Q}$. All coordinates zero also contradicts $\delta > 0$. This proves the claim.

1.6 A different rational bridge, with exact inverse and denominators

The preceding nonintersection concerns the specified map, not every map between the quadratic surfaces. The following construction uses projection from the rational point $(1,0,0)$ on $\Delta = -1$; it does not replace F or the physical meaning, if any, assigned to its coordinates.

Lemma 1.4. *The equation $\Delta(q) = 0$ has no nonzero rational solution.*

Proof. After clearing denominators use the integral S, U, V just defined. Expansion gives $2\Delta = 2S^2 - 3U^2 - V^2$. Modulo three, $2S^2 = V^2$ forces $3 \mid S, V$. Substitution then forces $3 \mid U$. Repeating gives divisibility of S, U, V by every power of three, hence all vanish. The linear map from q to (S, U, V) has nonzero determinant, so $q = 0$. □

Theorem 1.5. *For fixed $\delta \in \mathbb{Q}_{>0}$ write*

$$X_\delta = \{(a,b,c) : ac - b^2 = -\delta\}, \quad Y = \{q : \Delta(q) = -1\},$$

and define

$$D(a,b) = -1 - a^2 - b^2 + 4a + 4b + 4ab, \quad K(a,b) = 1 - 2a - 2b.$$

On $U_\delta = \{(a,b,c) \in X_\delta : aDK \neq 0\}$ the map

$$\mathcal{B}_\delta(a,b,c) = \left(1 + \frac{2K}{D}, \frac{2Ka}{D}, \frac{2Kb}{D}\right)$$

is an isomorphism of open varieties over $\mathbb{Q}$ onto the open part of Y where $q_1 \neq 1$, $q_2 \neq 0$, and $D(q_2/(q_1-1), q_3/(q_1-1)) \neq 0$. Its inverse is

$$a = \frac{q_2}{q_1 - 1}, \qquad b = \frac{q_3}{q_1 - 1}, \qquad c = \frac{b^2 - \delta}{a}.$$

Every integral matrix on X_δ with $a > 0$ belongs to U_δ. For such a matrix, the exact least common denominator of its image is

$$d_{\text{image}} = \frac{|D(a,b)|}{\gcd(|D(a,b)|, |2K(a,b)|)}.$$

In particular its image is integral exactly when $D(a,b) \mid 2K(a,b)$.

Proof. Set $q = (1,0,0) + \lambda(1,a,b)$. Expansion, with no change of scale in Δ, gives

$$\Delta(q) = -1 - 2K\lambda + D\lambda^2.$$

The second intersection with Y is therefore $\lambda = 2K/D$, nonzero on the stated domain. The displayed image and inverse follow. Conversely, for a point of the stated target open set, put $\lambda = q_1 - 1$ and recover a, b. Its equation gives $\lambda D = 2K$, so $K \neq 0$ and the constructions are inverse regular maps on the indicated opens. Recovering c retains the fixed determinant $-\delta$. For integral a, b, K is odd and cannot vanish. Moreover $D = \Delta(1,a,b) \neq 0$ by Lemma 1.4. Finally, $q_1 - 1 = 2K/D$ has reduced denominator $|D|/\gcd(|D|, |2K|)$, and q_2, q_3 are integer multiples of this fraction. This proves both the denominator formula and its exactness. □

For the packet's fixed opening $A_* = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\delta = 1$, $D = 9$, $K = -3$, and

$$\mathcal{B}_1(A_*) = (1/3, -2/3, -2/3), \qquad \Delta(1/3, -2/3, -2/3) = -1.$$

This rational image coexists with the irrationality of $\mathfrak{K}_1(A_*)$, because the two maps are different and both have now been specified. The rational bridge does not automatically preserve the integral positive chamber, the integral orbit count, or a dynamics. The denominator condition above is the exact next arithmetic restriction, and the inverse retains δ as a parameter rather than erasing it.

1.7 Independent checks and reproducibility

The separate verifier `supporting_materials/workbench/evidence/koide_es_v3_audit_20260904/verify_independent.py` enumerates the smaller denominator $N/R < y \le 2N/R$, then reconstructs $z = Ny/(Ry - N)$, independently of the packet's divisor enumerator. It agrees with all 1430 oriented states in 665 coprime shells through $n = 161$. It checks the 819 diagonal certificates for 329 primes below 5000, the symbolic rational-map identity and inverse, and 820 exact denominator cases. These finite regressions supplement, rather than replace, the proofs above. The original packet's verifier is replayed separately without modifying its source or its original output. The archive SHA-256 is

`ab5c21cf00268f9019c95ee24b565d05c4d9337c51e4a576ded9e3cc3f0f235e`

The correct full hash is recorded byte-for-byte in the evidence manifest; the line breaking here is typographical only.

2 The fifteen $p = 1201$ selector rays inside $\mathrm{Herm}_3(\mathbb{H})$

The common integer 15 in the complete $p = 1201$ selector census and in $\dim_{\mathbb{R}} \mathrm{Herm}_3(\mathbb{H}) = 15$ admits an exact test. The test gives a nontrivial morphism, but it rules out the interpretation of the selector states as fifteen independent Jordan-algebra directions.

2.1 Marked selectors and their prime families

Fix $p_0 = 1201$. A marked state in the complete census is a tuple

$$s = (\chi, R, h, r, t), \qquad \chi \in \{E, M\}, \qquad p_0 = 4rth - R, \qquad \gcd(r, t) = 1,$$

with positive integral R, h, r, t. Its channel gate is

$$\begin{array}{ll} E: & R \mid 4r^2h + 1, \\ M: & R \mid r + t. \end{array} \tag{1}$$

In the exterior channel the first gate is equivalent to $R \mid p_0 r + t$, because

$$p_0 r + t = t(4r^2h + 1) - Rr.$$

Put

$$(u, a, v) = (hr^2, hrt, ht^2), \qquad uv = a^2, \qquad R = 4a - p_0. \tag{2}$$

If $\kappa_E = (p_0 r + t)/R$ and $m = (r + t)/R$, the ordered denominator maps are

$$D_E(s) = (hrt,\ ht\kappa_E,\ p_0 hr\kappa_E), \tag{3}$$

$$D_M(s) = (hrt,\ p_0 hrm,\ p_0 htm). \tag{4}$$

Clearing the full common denominator proves

$$\sum_{d \in D_\chi(s)} \frac{1}{d} = \frac{4}{p_0}.$$

The fifteen states, with orientation and scale retained, are as follows. The integer L is the least prime increment that also preserves $p \equiv 361 \pmod{840}$.

tag	χ	R	h	(r,t)	u	a	v	L	Koide +
E23a	E	23	17	$(1,18)$	17	306	5508	57960	yes
M23a	M	23	3	$(6,17)$	108	306	867	14280	no
M23b	M	23	3	$(17,6)$	867	306	108	14280	no
E23b	E	23	34	$(9,1)$	2754	306	34	57960	no
M31a	M	31	2	$(1,154)$	2	308	47432	9240	yes
E31	E	31	77	$(4,1)$	1232	308	77	52080	no
M31b	M	31	2	$(154,1)$	47432	308	2	9240	no
M39a	M	39	2	$(1,155)$	2	310	48050	26040	yes
E39	E	39	62	$(5,1)$	1550	310	62	10920	no
M39b	M	39	2	$(155,1)$	48050	310	2	26040	no
M47a	M	47	1	$(8,39)$	64	312	1521	43680	no
M47b	M	47	1	$(39,8)$	1521	312	64	43680	no
E71	E	71	53	$(1,6)$	53	318	1908	59640	no
E339	E	339	11	$(35,1)$	13475	385	11	94920	no
E1043	E	1043	17	$(33,1)$	18513	561	17	1376760	no

For an exterior row define

$$h_k = h_0 + Rk, \qquad p_k = p_0 + 4rtRk;$$

for a middle row define

$$h_k = h_0 + k, \qquad p_k = p_0 + 4rtk.$$

The corresponding gate in (1) is preserved. The exterior prime class modulo $4rtR$ and the middle prime class modulo $4rt$ are reduced. Their least intersections with 361 (mod 840) are

$$L_s = \begin{cases} \operatorname{lcm}(4rtR, 840), & \chi = E, \\ \operatorname{lcm}(4rt, 840), & \chi = M. \end{cases} \tag{5}$$

Thus

$$p_n = 1201 + L_s n, \qquad h_n = h_0 + \frac{L_s}{4rt} n \tag{6}$$

retains the same marked ray, and Dirichlet's theorem gives infinitely many prime p_n in every row. The fifteen marked rows give eleven unordered denominator triples at p_0 and ten distinct residue classes 1201 (mod L_s). These are different quotients: middle reversal produces the first collapse, while E23a and E23b have distinct rays but the same $L_s = 57960$.

2.2 The exact Koide cone map

Define

$$\Gamma(s) = \begin{pmatrix} u & a \\ a & v \end{pmatrix} = h \begin{pmatrix} r \\ t \end{pmatrix} (r \ t). \tag{7}$$

This matrix is positive semidefinite of rank one and has Smith normal form $\operatorname{diag}(h, 0)$. The Lorentz-coordinate map

$$\mathcal{L}(\Gamma) = \left(\frac{u+v}{2}, a, \frac{u-v}{2} \right) = (H, X, Y) \tag{8}$$

satisfies $H^2 - X^2 - Y^2 = uv - a^2 = 0$.

Let

$$F(q_1, q_2, q_3) = \left(\frac{q_1 + q_2 + q_3}{\sqrt{3}}, \frac{q_1 - q_3}{\sqrt{2}}, \frac{q_1 + q_3 - 2q_2}{\sqrt{6}} \right).$$

This is orthogonal. Its inverse at (8) is

$$q_1 = \frac{u+v}{2\sqrt{3}} + \frac{a}{\sqrt{2}} + \frac{u-v}{2\sqrt{6}},$$

$$q_2 = \frac{u+v}{2\sqrt{3}} - \frac{u-v}{\sqrt{6}},$$
$$q_3 = \frac{u+v}{2\sqrt{3}} - \frac{a}{\sqrt{2}} + \frac{u-v}{2\sqrt{6}}. \tag{9}$$

For

$$Q_K(q) = \frac{2}{3}(q_1+q_2+q_3)^2 - (q_1^2+q_2^2+q_3^2)$$

orthogonality gives $Q_K(q) = H^2 - X^2 - Y^2 = 0$. Hence (9) is an exact coefficient-preserving map over $\mathbb{Q}(\sqrt{2},\sqrt{3})$, not a numerical fit. In the displayed orientation exactly E23a, M31a, and M39a lie in the positive Koide chamber.

2.3 Embedding and rank obstruction in the quaternionic Jordan algebra

Let $J = \mathrm{Herm}_3(\mathbb{H})$ with $A \circ B = (AB+BA)/2$, and put $e = \mathrm{diag}(1,1,0)$. Define

$$\iota : \mathrm{Sym}_2(\mathbb{R}) \longrightarrow J, \qquad \iota\begin{pmatrix} u & a \\ a & v \end{pmatrix} = \begin{pmatrix} u & a & 0 \\ a & v & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{10}$$

Block multiplication proves $\iota(A \circ B) = \iota(A) \circ \iota(B)$. Thus ι is an injective real-linear Jordan homomorphism with image

$$C_{\mathbb{R}} := \iota(\mathrm{Sym}_2(\mathbb{R})) = eJe \cap \mathrm{Herm}_3(\mathbb{R}), \qquad \dim_{\mathbb{R}} C_{\mathbb{R}} = 3.$$

The full Peirce corner is $eJe \cong \mathrm{Herm}_2(\mathbb{H})$ of real dimension 6; $C_{\mathbb{R}}$ is its proper real-symmetric Jordan subalgebra. It preserves trace, trace pairing, positivity, and ordinary rank. In the Moore convention

$$\begin{pmatrix} d_1 & q_3 & \bar{q}_2 \\ \bar{q}_3 & d_2 & q_1 \\ q_2 & \bar{q}_1 & d_3 \end{pmatrix},$$

the only nonzero coordinates of $\iota\Gamma(s)$ are $d_1 = u, d_2 = v, q_3 = a \in \mathbb{R}$. Its Moore cubic is zero.

Projectivizing the scale gives the idempotent

$$P_{r,t} = \frac{1}{r^2+t^2}\begin{pmatrix} r^2 & rt & 0 \\ rt & t^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad P_{r,t}^2 = P_{r,t}, \quad \mathrm{tr}\, P_{r,t} = 1. \tag{11}$$

It represents the quaternionic line $[r:t:0]_{\mathbb{H}}$. The fifteen rows have pairwise distinct ratios $[r:t]$, so the composition

$$\Sigma_{1201} \hookrightarrow \mathbb{RP}^1 \hookrightarrow \mathbb{HP}^2, \qquad s \longmapsto [r:t] \longmapsto [r:t:0]_{\mathbb{H}} \tag{12}$$

is injective.

Theorem 2.1 (Exact span and Jordan closure)**.** *The fifteen matrices $\iota\Gamma(s)$ are fifteen distinct projective rank-one points, but*

$$\mathrm{span}_{\mathbb{R}}\{\iota\Gamma(s) : s \in \Sigma_{1201}\} = \langle \iota\Gamma(\Sigma_{1201})\rangle_{\mathrm{Jordan}} = C_{\mathbb{R}}.$$

In particular, both spaces have real dimension 3*, not* 15*.*

Proof. In coordinates $\left(\begin{smallmatrix} x & y \\ y & z \end{smallmatrix}\right) \leftrightarrow (x, y, z)$, the projective image of (r, t) is (r^2, rt, t^2). The rows E31, E39, and E339 give

$$\begin{pmatrix} 16 & 4 & 1 \\ 25 & 5 & 1 \\ 1225 & 35 & 1 \end{pmatrix}, \qquad \det = -930 \neq 0.$$

Hence the images span the whole three-dimensional real-symmetric corner $C_{\mathbb{R}}$. The space $C_{\mathbb{R}}$ is a Jordan subalgebra by the block-multiplication identity above, so the generated Jordan algebra is contained in $C_{\mathbb{R}}$. A Jordan subalgebra is a real vector space; because it contains the fifteen generators, it contains their span $C_{\mathbb{R}}$. Both containments are equalities. □

The number fifteen is not invariant even after the hard residue class is held fixed. Complete divisor enumeration gives the following exact comparison:

p	1201	4561	8761	9601
$\#E$	7	26	16	17
$\#M$	8	28	20	12
$\#(E \sqcup M)$	15	54	36	29
#{unordered denominator triples}	11	40	26	23

All four displayed primes satisfy $p \equiv 361 \pmod{840}$. Thus neither the hard residue class nor the selector construction singles out a fifteen-element family; the equality $\#\Sigma_{1201} = \dim_{\mathbb{R}} \mathrm{Herm}_3(\mathbb{H}) = 15$ is a feature of the individual prime 1201, not an invariant of the arithmetic family. The deterministic enumeration and primality checks are included in the accompanying verification script.

At fixed p_0 the unprojectivized embedding preserves the markings. From $\iota\Gamma$ one recovers (u, a, v) and then

$$h = \gcd(u, a, v), \quad r = \sqrt{u/h}, \quad t = \sqrt{v/h}, \quad R = 4a - 1201, \quad j = (R - 3)/4.$$

The two channel gates cannot overlap in this census: overlap would imply $R \mid 1200$, and none of $23, 31, 39, 47, 71, 339, 1043$ divides 1200. Projectivization in (11) loses h, and without the fixed prime the residual formula $R = 4a - p$ is unavailable.

2.4 Symmetry and incidence obstructions

Normalized orientation reversal fixes the seven exterior states and swaps the four middle pairs, so its cycle type is $1^7 2^4$. The transvection

$$\tau(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_2, x_3, x_4)$$

on $\mathbb{F}_2^4 \setminus \{0\}$ has the same cycle type. For example, with integers $1, \ldots, 15$ encoding the four binary coordinates from the low bit, the following map is C_2-equivariant:

s	$E23a$	$E23b$	$E31$	$E39$	$E71$	$E339$	$E1043$
$\phi(s)$	1	4	5	8	9	12	13

s	$M23a$	$M23b$	$M31a$	$M31b$
$\phi(s)$	2	3	6	7

s	$M39a$	$M39b$	$M47a$	$M47b$
$\phi(s)$	10	11	14	15

This proves only an equivalence of C_2-sets. It is noncanonical: a set with this cycle type has $7!2^4 4! = 1{,}935{,}360$ automorphisms commuting with the involution.

It cannot be upgraded to the natural projective incidence. The selector points lie on the nonsingular conic $xz - y^2 = 0$ in $\mathbb{P}^2(\mathbb{R})$, so no three distinct selector images are collinear. By contrast, $PG(3, 2)$ has 35 three-point lines. Nor does normalized reversal extend linearly on the canonical real-symmetric corner: it fixes E31, E39, and E339, which span $C_{\mathbb{R}}$, so such a linear map would be the identity; but it must exchange four nonidentical middle pairs.

The other elementary count comparisons do not supply a bridge. Type A_3 has 12 oriented roots, 6 unoriented root lines, and $4 + 6 + 4 = 14$ nonzero minuscule weights. The rank-one idempotents of J form the real eight-dimensional continuum $\mathbb{HP}^2$, while the present images occupy the proper copy $\mathbb{RP}^1$ in (12).

2.5 Conclusion and next exact test

The current structural statement is therefore

$$\boxed{\begin{array}{c}\text{fifteen marked prime-carrying selector rays } \hookrightarrow \mathbb{RP}^1 \hookrightarrow \mathbb{HP}^2, \\ \dim \text{span} = \dim \text{Jordan closure} = 3.\end{array}}$$

The equality between the number of marked rays and $\dim_{\mathbb{R}} J$ is not explained by the defined morphism.

A decisive higher-rung test would construct, from the arithmetic residual and channel markings, an explicit element g_s of a named quaternionic structure group and study

$$g_s \iota(\Gamma(s)) g_s^*.$$

One must then compute the span and Jordan closure over complete multi-prime censuses and prove equivariance with the selector involutions. If all images remain in one common three-dimensional real-symmetric Jordan corner, the obstruction persists; if the markings canonically move them through enough conjugate corners and the closure is all of J, that would be a genuine fifteen-dimensional bridge. No such marking-to-conjugation map is presently defined.